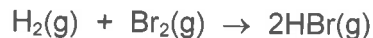


- 13** In reaction 1, hydrogen reacts with gaseous bromine to form hydrogen bromide.



In reaction 2, hydrogen reacts with gaseous iodine to form hydrogen iodide.



For reaction 1, the rate equation is

$$\text{rate} = \frac{k_1[\text{H}_2][\text{Br}_2]^{1.5}}{[\text{Br}_2] + k_2[\text{HBr}]}$$

For reaction 2, the rate equation is

$$\text{rate} = k[\text{H}_2][\text{I}_2]$$

What is correct about these reactions?

- 1 For the hydrogen and bromine reaction, the formation of HBr slows down the rate of the reaction.
- 2 Only the hydrogen and iodine reaction could be a single step reaction.
- 3 Doubling the concentration of the halogen in each reaction doubles each rate of reaction.

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 2 and 3 only      **D** 1 only